Customer Feedback Sentiment Analysis using BERT

Luis Lopez-Echeto Bordon
UCSC Extension Course: AISV.X401.(21)
Summer 2025

Problem Statement

The Challenge

Businesses struggle to efficiently process vast amounts of customer feedback from surveys and reviews.

The Goal

Develop an automated system to quickly classify customer feedback as positive, or negative.

Impact

Help companies understand customer sentiment faster to improve products and services.

Problem Statement: Why BERT?

Contextual Understanding: BERT grasps the full meaning of a sentence, avoiding the pitfalls of older models.

Highly Accurate: Its deep learning architecture provides reliable business insights.

Efficient: By using **transfer learning**, we can fine-tune a pre-trained model with a small amount of data.

Project: Data Preprocessing

Dataset: Customer feedback from Kaggle.

Challenge: Data used a non-standard format with quotes wrapping each row and escaped quotes within the text.

```
csv
"Text, Sentiment, Source, Date/Time, User ID, Location, Confidence Score"
"""I love this product!"", Positive, Twitter, 2023-06-15 09:23:14, @user123, New York, 0.85"
```

Solution: Built a custom parser to clean the text.

Outcome: The data structured as a pandas DataFrame, ready for analysis and modeling.

Preparing Data for the BERT Model

- **Label Encoding:** Convert text sentiment labels into a numerical format.
  - **Positive** reviews were mapped to 1.
  - **Negative** reviews were mapped to 0.

Text	Sentiment	label
I love this product!	Positive	1
The service was terrible.	Negative	0
This movie is amazing!	Positive	1

- **Minimal Text Preprocessing:** We only stripped extra whitespace.
- **Why so little preprocessing?**
  - BERT handles tasks like lowercasing, punctuation removal, and stemming with its own internal tokenizer.
  - This simplifies our pipeline and preserves valuable context for the model.

Preparing Data for Training

- **Split the Data**: We divided the dataset into an 80% training set and a 20% validation set. The training set teaches the model; the validation set tests it on new data.
- **Tokenization**: The `BertTokenizer` converts raw text into a numerical format. It also handles padding (making sentences the same length) and truncation (cutting long sentences).
- **Prepare for Pytorch**: A custom `torch.utils.data.Dataset` class was created to correctly format the tokenized data for training in batches.

Model Training and Evaluation

- **The Process**: We fine-tune a pre-trained BERT model for our task. We define key parameters (epochs, batch size, etc.) using `TrainingArguments`.
- **The `Trainer`**: This class from Hugging Face automates the entire training loop. It efficiently handles all the complex steps, including data batching, calculating the loss, backpropagation, and updating the model's weights.
- **Evaluation**: After training, the model is evaluated on a separate validation set. This ensures we have an unbiased measure of its performance on new, unseen data.
- **Result**: The final result is a model ready to make fast and accurate sentiment predictions.

Examples in Action: Positive Feedback

`"I'm honestly impressed – I thought this would take days."`

Model Prediction: Negative (0.56 confidence)

Why? The model misread the slightly sarcastic or cynical tone, which highlights the need for a human-in-the-loop system to validate nuanced language.

`"It’s working now."`

Model Prediction: Positive (0.85 confidence)

Why? The model assigned high confidence to a brief, positive-sounding statement, even though the user was likely just stating a fact. This reminds us that context is key for short, simple replies.

Examples in Action: Negative Feedback

Negative Feedback:

`"Please escalate this – I’ve emailed three times and still no reply."`

Model Prediction: Negative (0.73 confidence)

Why? The model correctly flagged frustration and urgency based on keywords like "escalate" and phrases indicating repeated attempts without a solution. This is ideal for an automated triage system.

`"Still no resolution. Starting to regret choosing your platform."`

Model Prediction: Negative (0.86 confidence)

Why? The model correctly identified a high degree of negative sentiment, indicating a user who is likely to churn. This is critical for customer retention.

Project Summary

- **The Solution:** We built a robust sentiment analysis system by fine-tuning a pre-trained **BERT model**.
- **The Outcome:** The model transforms unstructured customer feedback into valuable, actionable business insights with high accuracy.

- **The Limitation:** Our model, while powerful, can still struggle with complex human nuances like sarcasm.
- **The Best Approach:** The ideal solution is a **"human-in-the-loop"** system, where the AI flags ambiguous feedback for a human to review, ensuring complete accuracy.

Thanks! 🎉

Github: elchicha